Probabilistic Combinatorics

Sheet 11

July 27, 2024

Problem 1. There are 799 teams in a tournament. Here every team plays against every other team exactly once and every match is decisive, i.e. there are no ties. Show that there are 2 distinct sets A and B of 7 teams each, such that every team of A has won against every team of B.

Proof.
This problem can be modelled by a tournament $G = (V, E)$ where a vertex $v \in V$ presents a unique team and an edge $e = (u, v) \in E$, where $u, v \in V$, presents the event where the team presented by the vertex u has won the team presented by the vertex v. That is, the original problem turns into the problem of showing that there exist two distinct sets of vertices in G, which are named A and B, such that for every vertex in A, the vertex must have an outgoing edge to every vertex in B.

Let $(\Omega, \mathbb{P})$ be the discrete probability space where the sample space Ω is the set of choices of 7 vertices from G for set A and the probability measure $\mathbb{P}(w) = 1/\binom{799}{7}$ for any $w \in \Omega$. Let $X : \Omega \to \mathbb{R}$ be the random variable that assigns to $w \in \Omega$ number of vertices in G having an incoming edge from each vertex in ω. For any vertex v in G, we define the random variable $X_v : \Omega \to \mathbb{R}$ defined by the following rule for any $\omega \in \Omega$

$$X_v(w) = \begin{cases} 1 \text{ if } v \text{ has an incoming edge from each vertex in } w \\ 0 \text{ otherwise} \end{cases}$$

Next, we define the indicator random variable $X_v = 1_{A_v}$ of event

$$A_v = \{w \in \Omega \mid v \text{ has an incoming edge from each vertex in } w\}$$

then $\mathbb{E}(X_v) = \mathbb{E}(1_{A_v}) = \mathbb{P}(A_v)$. For each vertex $v \in V$, it is obvious that $|A_v| = \binom{deg^- v}{7}$. And thus

$$\mathbb{P}(A_v) = \binom{deg^- v}{7} / \binom{799}{7}$$

Note that $X = \sum_{v \in V} X_v$. By linearity of expectation, we have

$$\mathbb{E}(X) = \sum_{v \in V} \mathbb{E}(X_v) = \sum_{v \in V} \mathbb{P}(A_v) = (\sum_{v \in V} \binom{deg^- v}{7}) / \binom{799}{7} \tag{1.1}$$

It is worth to note that, when $deg^- v < 7$ for any $v \in V$, then $\binom{deg^- v}{7} = 0$. That is, only the vertices $v \in V$ having $deg^- v \geq 7$ are important to the sum $\sum_{v \in V} \binom{deg^- v}{7}$. For any vertex $v \in V$ such that $deg^- v \geq 7$, let

$$f(deg^- v) = \binom{deg^- v}{7}$$

It is known that $\binom{a}{b}$ is a convex function for $a \geq b$, thus $f(deg^- v)$ is a convex function. Therefore, by Jensen's inequality, we have

$$\begin{aligned} & f(\frac{1}{|V|} \sum_{v \in V} deg^- v) \leq \frac{1}{|V|} \sum_{v \in V} f(deg^- v) \\ \Leftrightarrow\ & |V| f(\frac{1}{|V|} \sum_{v \in V} deg^- v) \leq \sum_{v \in V} f(deg^- v) \end{aligned} \tag{1.2}$$

It is known that $\sum_{v\in V} deg^- v$ is equal to number of edges in G, thus $\sum_{v\in V} deg^- v = \binom{799}{2}$ since G is a tournament. Therefore, the inequality (1.2) becomes

$$\begin{aligned}
&\sum_{v\in V} f(deg^- v) \geq 799 f\left(\frac{1}{799}\binom{799}{2}\right) \\
\Leftrightarrow &\sum_{v\in V} f(deg^- v) \geq 799 f(399) \\
\Leftrightarrow &\sum_{v\in V} f(deg^- v) \geq 799\binom{399}{7} \qquad (1.3)
\end{aligned}$$

Based on the equation (1.3) and the definition of the function $f(deg^- v)$, the equation (1.1) is equivalent to

$$\mathbb{E}(X) \geq \left(799\binom{399}{7}\right) / \binom{799}{7} \Leftrightarrow \mathbb{E}(X) \geq 6.02$$

Since X only takes positive integers, the inequality above implies that $\mathbb{P}(X \geq 7) > 0$. Therefore, there must exists a set of 7 vertices in G where each vertex has an incoming edge from each vertex in A. Let this set be B, it is easy to verify that B must be distinct from A as there are no loops in a tournament, and the proof is finished. □

Problem 2. A proper vertex colouring of an undirected graph $G = (V, E)$ with a set of colours F is a map $f : V \to F$, such that for every edge $e \in E$ we have $\#f(e) = 1$, i.e. no adjacent vertices are coloured the same way. Let $G = ([n], e)$ be a bipartite graph and F a finite set of colours. Let $F_1, F_2, \ldots, F_n \in F$, such that $\#F_i \geq \log_2(n)$ for $1 \leq i \leq n$. Show that there is a proper vertex colouring of G such that $f(i) \in F_i$ for every $1 \leq i \leq n$.

Proof.
Since G is bipartite, then its vertices can be partitioned into two disjoint sets, which are named A and B, such that the edges in the graph connect a vertex in A to a vertex in B or vice-versa. That is, if we divide the set of colours F into two disjoint sets F_A and F_B, and only use colours F_A and F_B to colour vertices in A and B respectively, then such colouring must be a proper colouring. Additionally, we want to ensure that $f_i \in F_i$ for every $1 \leq i \leq n$ when we colour G in this way. That is, for any partition of F into F_A and F_B, we define a vertex numbered i in A or B is said to be colourable only when $F_i \cap F_A \neq \emptyset$ or $F_i \cap F_B \neq \emptyset$, respectively. Therefore, the original problem turns into the problem of showing that there exists a way to partition the set F into two disjoint sets F_A and F_B such that all vertices in G are colourable.

Firstly, we define the discrete probability space $(\Omega, \mathbb{P})$ where the sample space Ω is the set of all possible partitions of F into two disjoint sets F_A and F_B, and the probability measure $\mathbb{P}(w) = (\frac{1}{2})^{|F|}$ for any $\omega \in \Omega$. Fixed a vertex $v \in V$, let

$$\Omega_v = \{\omega \in \Omega \mid \ v \text{ is not colourable}\}$$

To be more specific, Ω_v is the set of partitions of F into two disjoint sets F_A and F_B where all colours in F_v are distributed to the wrong set. Consider any colour in F_v, the probability of the colour to be distributed to the wrong set is $\frac{1}{2}$, as there are only two sets: F_A and F_B. Additionally, there are at least $\log_2(n)$ colours in F_v. Therefore,

$$\mathbb{P}(\Omega_v) \leq \left(\frac{1}{2}\right)^{\log_2(n)} = \frac{1}{n}$$

Furthermore, we have

$$\begin{aligned}
&\mathbb{P}(\bigcup_{v\in V} \Omega_v) \leq \sum_{v\in V} \mathbb{P}(\Omega_v) \\
\Leftrightarrow &\mathbb{P}(\bigcup_{v\in V} \Omega_v) \leq n\frac{1}{n} \\
\Leftrightarrow &\mathbb{P}(\bigcup_{v\in V} \Omega_v) \leq 1 \qquad (2.1)
\end{aligned}$$

Moreover, the equal sign in the inequality (2.1) only happens when $\Omega_u \cap \Omega_v = \emptyset$ for any distinct vertices $u, v \in V$. This implies that, for any partition of F into two disjoint sets F_A and F_B such that a vertex $u \in V$ is non-colourable, then other vertices in V that are not u must be colourable. It is easy to verify that this is impossible. Hence, it is impossible for the equal sign in the inequality (2.1) to happen, thus

$$\mathbb{P}(\bigcup_{v \in V} \Omega_v) < 1$$

This inequality implies that not every partition of F into two disjoint sets F_A and F_B that contain at least one vertex in G that is non-colourable. Therefore, there must exist a partition of F into two disjoint sets F_A and F_B where every vertex in G is colourable, and thus there exists a proper vertex colouring of G such that $f(i) \in F_i$ for every $1 \leq i \leq n$ due to the logic presented at the beginning of the proof. □

Problem 3. There is a chess tournament with 40 players. There are 80 matches played and no two players play against each other more than once. Show that there are at least 5 players, where no two of them play against each other.

Proof.
The chess tournament can be modelled by an undirected graph $G = (V, E)$ where a vertex presents a unique player while an edge between two vertices presents a match between two players. Therefore, $|V| = 40$ and $|E| = 80$ and the original problem turns into the problem of showing that the graph G has an independent set of size 5 at least.

Recall the Theorem 4.1 in the lecture that the maximum cardinality of an independent set in G is

$$\alpha(G) \geq \frac{n}{2d}$$

where $|V| = n$ and $|E| = \lfloor \frac{nd}{2} \rfloor$. In this case, we have $d = 4$, and thus $\alpha(G) \geq 5$ based on the Theorem 4.1, and the proof is finished. □

Problem 4. Let $(a_n)_{n \in \mathbb{N}}$ and $(b_n)_{n \in \mathbb{N}}$ be sequences of real numbers. Decide which of the following statements is always true , which might occur and which can't be true. Give a corresponding proof (for "is always true" or "can't be true") or examples (one example for "holds" and for "doesn't hold" for each "might occur").

(i) $a_n = O(a_n^2)$

(ii) $a_n \neq O_{b_n}$ and $b_n \neq O(a_n)$

(iii) $a_n = O(b_n)$ and $b_n = o(a_n)$

Proof.
We are going to solve the problems one after another

(i) This case might occur. When $a_n = n$ for $n \geq 1$ then $a_n = O(a_n^2)$. To be more specific, we have

$$\limsup_{n \to \infty} \frac{|a_n|}{|a_n^2|} = \limsup_{n \to \infty} \frac{1}{|a_n|} = \limsup_{n \to \infty} \frac{1}{|n|} = 0$$

On the other hand, when $a_n = \frac{1}{n}$ for $n \geq 1$ then $a_n \neq O(a_n^2)$. This is because

$$\limsup_{n \to \infty} \frac{|a_n|}{|a_n^2|} = \limsup_{n \to \infty} \frac{1}{|a_n|} = \limsup_{n \to \infty} |n| = \infty$$

□